

Week 1: ArcIris genuine/impostor analysis on NDCVRL_002

Summary of work, findings and key points

Key points

- ArcIris (pretrained, *no retraining*) gives $d' = 3.42$ on 5,169 images / 248 subject-eyes. Impostor scores centre at 1.566 rad (near $\pi/2$), genuine at 0.837.
- At the 1% impostor cut (1.406 rad), 130,228 of 13.3M impostor pairs (0.98%) are false matches, and 5.4% of genuine pairs are missed.
- ArcIris has **no eyelid mask**. Occlusion strongly causes *false non-matches* (genuine tail) but barely affects the *impostor* tail.
- The impostor tail has no single cause. The clearest identified factor is the **960×640 wide-field capture format** (sensors `nd1N00059`, `nd1N00073`): small, low-texture irises that false-match at 5.25% vs 0.83% for 640×480 pairs.
- Glasses, contacts, eye colour and eyelid layout are weak or absent effects. Same-person opposite-eye pairs are enriched in the tail (5×) but are correctly labelled impostors.

1 Setup

- **Method:** ArcIris (`OpenSourceIrisRecognition`, `cfg_baseline`, pretrained weights). Score = arccos of cosine similarity of 512-d embeddings, radians, *lower = more similar*. All models run in `eval/inference` mode; nothing was trained or fine-tuned.
- **Data:** NDCVRL_002, 5,175 images, manifest built from `recording_metadata.csv` (all images matched). 5,169 passed `checkQuality`.
- **Pairs:** 70,876 genuine (same subject and eye) and impostor pairs (all others). The pipeline sampled 100,000 of the 13,285,820 impostor pairs; later analyses used all pairs.
- **Cut:** 1st percentile of sampled impostor scores, 1.4057 rad. A pilot on 100 images (20 subject-eyes) gave $d' = 4.58$ before the full run.

2 Results

2.1 Overall separation

Genuine mean 0.837, impostor mean 1.566, lowest impostor score 1.215, $d' = 3.42$; 5.4% of genuine pairs score above the cut.

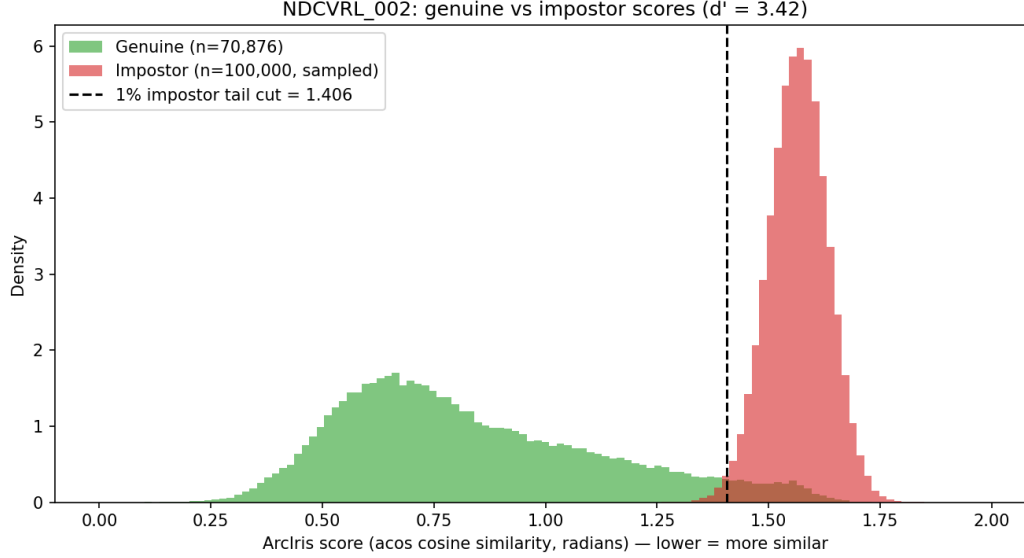


Figure 1: Genuine vs impostor score distributions (E0).

2.2 Embedding space

A t-SNE of the 512-d embeddings shows most identities as tight single-colour clusters, a few diffuse identities and a crowded central knot; left and right eyes are interleaved. PC1+PC2 explain only 3.4% of variance (rechecked in float64). Figure E1 in the evidence folder.

2.3 Eyelids and occlusion

- **Code:** `circApprox` predicts only pupil and iris circles; the whole ring is unwrapped and passed to the network. `checkQuality` never tests coverage (an image with 100% occlusion passed).
- **Test:** the dataset’s segmentation masks were unwrapped with ArcIris’s own circles; mean visible iris is 72.6%.
- **Genuine pairs (by occlusion of the more-occluded image):** miss rate is 0.33% (Q1), 0.91%, 2.59%, and **17.75% (Q4)** (59 vs 3,152 misses; ratio 53×, 95% cluster-bootstrap CI 28–144×).
- **Impostor pairs:** score is essentially unrelated to occlusion (Spearman -0.03); false-match rate 0.9% $\rightarrow$ 1.3% in the most-occluded quartile.
- **Masking without retraining** (blank occluded pixels to mid-grey, 900-image subset) *worsened* results: d' 3.73 $\rightarrow$ 3.13. The network never saw such input, so mask-aware use would need retraining (out of scope).

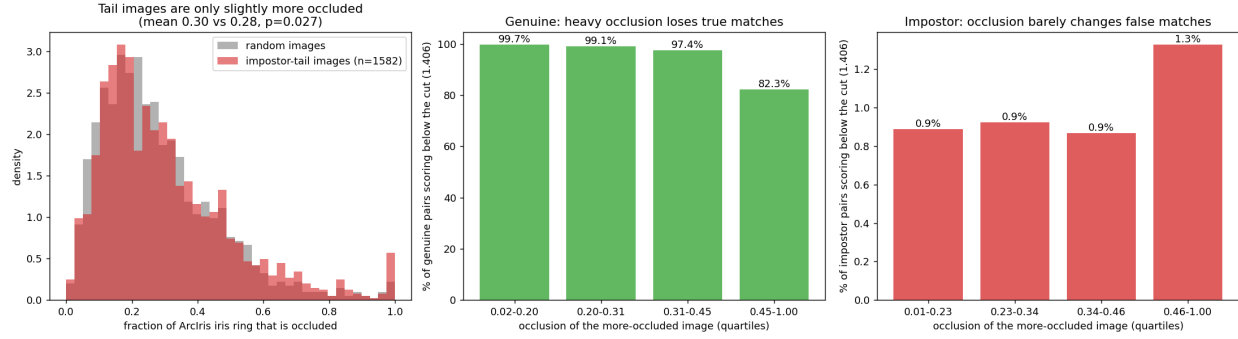


Figure 2: Occlusion vs misses (genuine) and false matches (impostor) (E2).

2.4 What the impostor false matches share

Among the 1,000 lowest-scoring sampled impostor pairs (1,582 images, 178 subjects):

- Same sensor 34.3% (vs 24.2% for all impostor pairs), same environment 28.3% (18.1%), same eye side 60.5% (50.2%).
- Hub identities: nd1S05271 L appears in 68 of the 1,000 pairs.
- 16 pairs (1.6%, vs 0.31% baseline) are the same person's opposite eyes. A label audit found all 32 images closer to their own labelled eye than to the other eye, and 0 of 5,169 images sit closer to another identity, so none is a hidden true match.

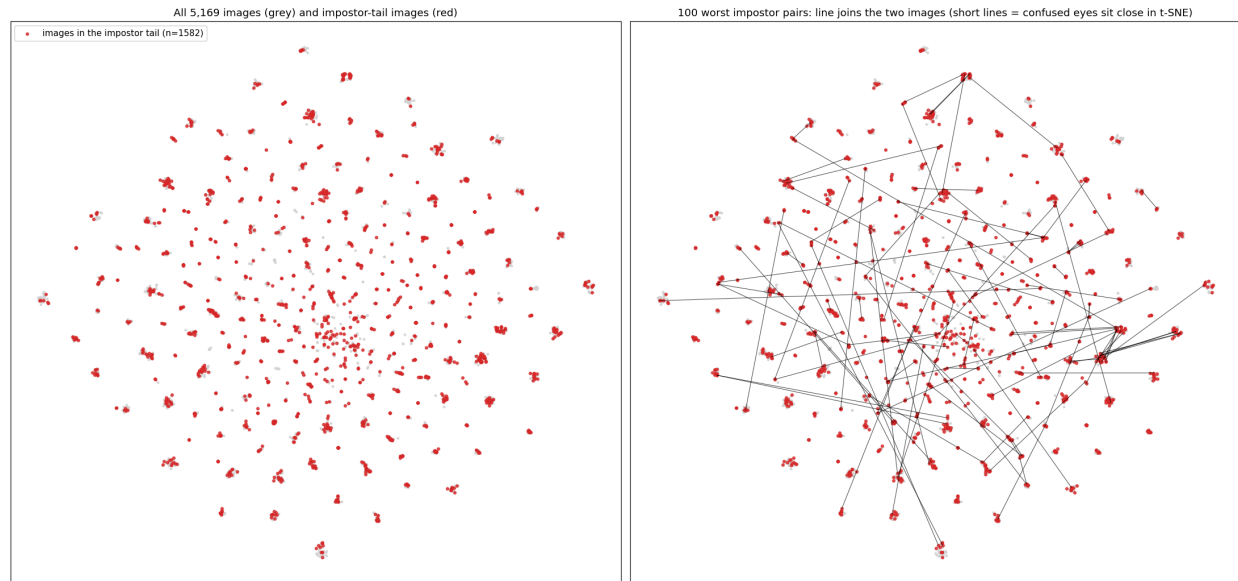


Figure 3: Impostor-tail images (red) on the t-SNE; right: 100 worst pairs joined (E6).

Tail identities have looser clusters (mean same-identity similarity 0.466 vs 0.524), and the tail is spread across essentially all identities rather than a few look-alike groups.

2.5 Format, texture and other factors (all 13.3M pairs)

- **Wide-field format:** 909 images (133 identity-eyes) are 960×640 , only from sensors 59 and 73. Both-wide pairs false-match at 5.25% ($5.4 \times$ baseline); 640×480 pairs 0.83%; mixed-format pairs 0.87%. Their irises are smaller (about 66 vs 119 px radius) and flatter (fine-texture 2.10 vs 5.27). Sensor and format are perfectly confounded.
- **Texture:** an apparent “low-texture pairs are worse” effect (2.99% vs 0.67–1.03%) is mostly this format: within 640×480 only, low-texture pairs 0.92% vs 0.88% (no effect).
- **Other attributes (both images share it):** glasses 1.43% ($1.45 \times$), contacts about baseline, eye colour only Black elevated ($2.0 \times$), eyelid-layout similarity flat (0.9–1.1%). Cosmetic contacts ($4.4 \times$) and some demographic groups rest on only 4–9 identity-eyes and are anecdotal.

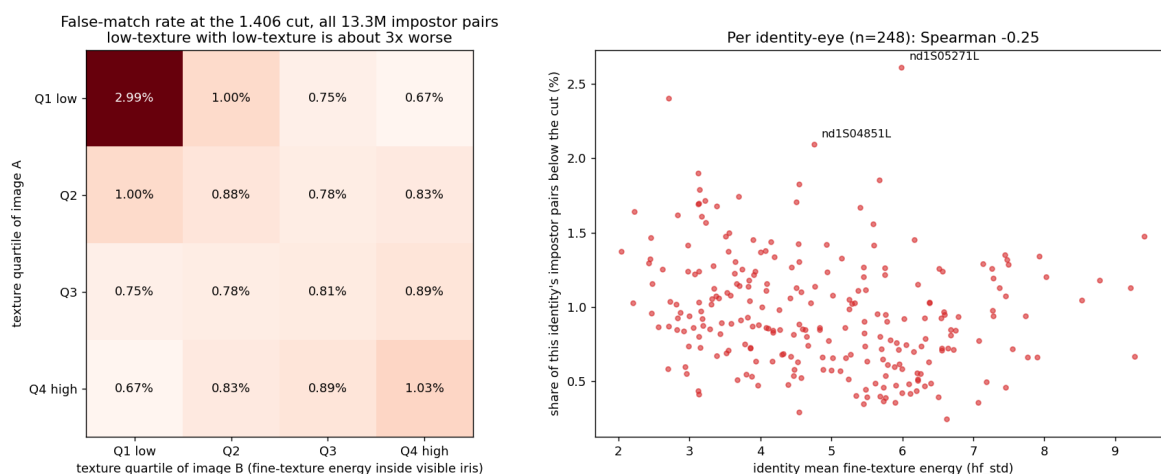


Figure 4: Texture vs false-match rate (E8). See the format caveat above.

2.6 Galleries

Side-by-side galleries were produced: 1,000 confused impostor pairs (`imposters/`, a ≈ 1 -in-130 sample of the 130,228 pairs under the cut), 10 genuine pairs at spread percentiles (`true matches/`), and the 16 same-person opposite-eye pairs. Examples: the worst impostor pair is a pair of dark wide-field frames; the missed genuine example shows a heavily lidded capture 14 months later.

3 Conclusions

1. ArcIris separates genuine from impostor well on clean 640×480 data (see Week 2: $d' = 4.40$), but the aggregate $d' = 3.42$ hides very different capture conditions.
2. **Occlusion (eyelids) drives false non-matches, not the impostor tail.** The long *right* tail of genuine scores is largely an occlusion effect.
3. The impostor tail is multi-factor: wide-field format (identified), capture-condition similarity (sensor, environment, eye side), hub identities, and natural spread. No single factor explains it.

4. Simple masking cannot fix occlusion without retraining, so any fix within “no retraining” must be at the score/threshold or quality-gate level.

4 Caveats

- Pairs share images and identities, so counts within subgroups are not independent; several subgroup effects rest on few identity-eyes.
- Sensor and image format are confounded; a resolution-vs-sensor control was not run.
- Occlusion is measured with the dataset’s segmentation masks; texture is one proxy (fine-texture energy).
- Findings come from one dataset and have not yet been confirmed on the held-out split; about 20 comparisons were run on the same data.

5 Files

- `Task 1 results/findings_log.md`: full log (F1–F13) with numbers and evidence.
- `Task 1 results/evidence/`: figures E0–E8.
- `arciris_analysis/`: scripts (manifest, pipeline, occlusion, masked re-scoring, texture, evidence).